

Question 1:

```
himasree@LAPTOP-41COT5AA:~/OSLab$ nano program02.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc program02.c -o program02
himasree@LAPTOP-41COT5AA:~/OSLab$ ./program02
Error opening source file: No such file or directory
Successfully opened/created 'destination.txt' with File Descriptor: 3
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {

    // 1. Open an existing file for reading
    int fd_read = open("source.txt", O_RDONLY);

    if (fd_read < 0) {
        perror("Error opening source file");
    } else {
        printf("Successfully opened 'source.txt' with File Descriptor: %d\n", fd_read);
        close(fd_read);
    }

    // 2. Open/Create a new file for writing with permissions (0644)
    int fd_write = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd_write < 0) {
        perror("Error creating destination file");
    } else {
        printf("Successfully opened/created 'destination.txt' with File Descriptor: %d\n", fd_write);
        close(fd_write);
    }
}
```

Question 2:

```
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc program23.c -o program23
himasree@LAPTOP-41COT5AA:~/OSLab$ ./program23
Read 33 bytes from file:
Hello World
This is a test file.
```

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

#define BUFFER_SIZE 128

int main() {

    int fd = open("source.txt", O_RDONLY);

    if (fd < 0) {
        perror("Error opening file to read");
        exit(EXIT_FAILURE);
    }

    char buffer[BUFFER_SIZE];

    // Read up to BUFFER_SIZE bytes from file descriptor into memory buffer
    ssize_t bytes_read = read(fd, buffer, sizeof(buffer) - 1);

    if (bytes_read < 0) {
        perror("Error reading file");
    } else {
        buffer[bytes_read] = '\0'; // Null-terminate string
        printf("Read %ld bytes from file:\n%s\n", bytes_read, buffer);
    }
}
```

```

// Read up to BUFFER_SIZE bytes from file descriptor into memory buffer
ssize_t bytes_read = read(fd, buffer, sizeof(buffer) - 1);

if (bytes_read < 0) {
    perror("Error reading file");
} else {
    buffer[bytes_read] = '\0'; // Null-terminate string
    printf("Read %ld bytes from file:\n%s\n", bytes_read, buffer);
}

close(fd);

return 0;

```

Question 3:

```

himasree@LAPTOP-41COT5AA:~/OSLab$ nano program24.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc program24.c -o program24
himasree@LAPTOP-41COT5AA:~/OSLab$ ./program24
Successfully wrote 68 bytes to 'destination.txt'

```

```

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {

    int fd = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("Error opening file to write");
        exit(EXIT_FAILURE);
    }

    char *text = "Hello! This text is written directly using the write() system call.\n";

    // Write buffer contents to file descriptor
    ssize_t bytes_written = write(fd, text, strlen(text));

    if (bytes_written < 0) {
        perror("Error writing to file");
    } else {
        printf("Successfully wrote %ld bytes to 'destination.txt'\n", bytes_written);
    }
}

```

Question 4:

```

himasree@LAPTOP-41COT5AA:~/OSLab$ gcc program25.c -o program25
himasree@LAPTOP-41COT5AA:~/OSLab$ ./program25
File opened successfully. File Descriptor ID: 3
File descriptor 3 closed successfully.

```

```

#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {

    int fd = open("source.txt", O_RDONLY);

    if (fd < 0) {
        perror("Failed to open file");
        exit(EXIT_FAILURE);
    }

    printf("File opened successfully. File Descriptor ID: %d\n", fd);

    // Explicitly closing the file descriptor releases kernel resources
    if (close(fd) == 0) {
        printf("File descriptor %d closed successfully.\n", fd);
    } else {
        perror("Error closing file descriptor");
    }

    return 0;
}

```

History:

```

himasree@LAPTOP-41COT5AA:~/OSLab$ history
1
2 pwd
3 ls
4 ~
5 cd ~
6 pwd
7 ls
8 mkdir test
9 cd test
10 pwd
11 wsl -l -v
12 its saying command not found for wsl -l -v
13 sudo apt update
14 sudo apt install build-essential
15 gcc --version
16 mkdir OSLab
17 gedit hello.c
18 sudo apt install gedit
19 gedit hello.c
20 gcc hello.c -o hello
21 gcc hello.c -o hello
22 ./hello
23 nano process.c
24 ls
25 gcc process -o process
26 ./process
27 gcc process.c -o process
28 ./process
29 ./process

```

```

30 sudo apt install calender
31 sudo apt update
32 sudo apt install calender
33 sudo apt install ncal
34 free -h
35 linux_command.c
36 gcc linux_command.c -o linux_command
37 echo
38 echo $ SHELL
39 echo $SHELL
40 fork_demo.c
41 mkdir OSLab cd OSLab
42 /OSLab
43 gedit fork_demo.c
44 cd /OSLab/OSLab
45 cd OSLab/OSLab
46 cd ~/OSLab/OSLab
47 ~/OSLab
48 cd ~/OSLab
49 nano program.c
50 gcc program.c -o
51 gcc program.c -o program
52 nano program.c
53 gcc program.c -o program
54 nano program.c
55 gcc program.c -o program
56 nano program.c
57 gcc program.c -o program
58 ./program
59 nano program.c

```

```

60 uname -a
61 lscpu
62 lscpu
63 lsblk
64 lsblk
65 ps -ef
66 top
67 nano exp1.c
68 gcc exp1.c -o exp1
69 ./exp1
70 nano exp2.c
71 gcc exp2.c -o exp2
72 ./exp2
73 nano exp3.c
74 gcc exp3.c -o exp3
75 ./exp3
76 nano exp4.c
77 gcc exp4.c -o exp4
78 ./exp4
79 nano exp5.c
80 gcc exp5.c -o exp5
81 ./exp5
82 nano exp6.c
83 gcc exp6.c -o exp6
84 ./exp6
85 nano exp7.c
86 gcc exp7.c -o exp7
87 ./exp7
88 nano exp8.c
89 gcc exp8.c -o exp8

```

```

90 ./exp8
91 nano exp9.c
92 gcc exp9.c -o exp9
93 ./exp9
94 nano exp1.c
95 nano exp2.c
96 nano exp3.c
97 nano exp4.c
98 cd ~/OSLab
99 history
100 cd ~/OSLab
101 explorer.exe .
102 Mkdir
103 mkdir
104 Rmdir
105 cp
106 gcc exp1.c -o exp1
107 ./exp1
108 gcc exp2.c -o exp2
109 ./exp2
110 gcc exp3.c -o exp3
111 ./exp3
112 gcc exp4.c -o exp4
113 ./exp4
114 gcc exp5.c -o exp5
115 ./exp5
116 gcc exp6.c -o exp6
117 ./exp6
118 gcc exp7.c -o exp7
119 ./exp7

```

```

120 gcc exp5.c -o exp5
121 ./exp5
122 cd ~/OSLab
123 gcc exp1.c -o exp1
124 ./exp1
125 gcc exp2.c -o exp2
126 ./exp2
127 gcc exp3.c -o exp3
128 ./exp3
129 gcc exp4.c -o exp4
130 ./exp4
131 gcc exp5.c -o exp5
132 ./exp5
133 gcc exp6.c -o exp6
134 ./exp6
135 gcc exp7.c -o exp7
136 ./exp7
137 gcc exp8.c -o exp8
138 ./exp8
139 nano exp9.c
140 gcc exp1.c -o exp1
141 cd ~/OSLab
142 gcc exp1.c -o exp1
143 ./exp1
144 gcc exp2.c -o exp2
145 ./exp2
146 gcc exp3.c -o exp3
147 ./exp3
148 gcc exp4.c -o exp4
149 ./exp4

```

```

150 gcc exp5.c -o exp5
151 ./exp5
152 gcc exp6.c -o exp6
153 ./exp6
154 gcc exp7.c -o exp7
155 ./exp7
156 gcc exp8.c -o exp8
157 ./exp8
158 gcc exp9.c -o exp9
159 ./exp.9
160 ./exp9
161 cd ~/OSLab
162 history
163 cd ~/OSLab
164 cd exec
165 mkdir exec
166 cd exec
167 nano programExec1.c
168 gcc programExec1.c -o programExec1
169 ./programExec1
170 nano programExec2.c
171 gcc programExec2.c -o programExec2
172 ./programExec2
173 nano programExec3.c
174 gcc programExec3.c -o programExec3
175 ./programExec3
176 nano programExec4.c
177 gcc programExec4.c -o programExec4
178 ./programExec4
179 nano programExec5.c

```

```

180 gcc programExec5.c -o programExec5
181 ./programExec5
182 cd ~/OSLab
183 pwd
184 ls
185 cd
186 mkdir
187 rmdir
188 cp
189 cd ~/OSLab
190 mkdir
191 mv
192 rm
193 cat
194 find
195 history
196 shell usage
197 cd ~/OSLab
198 mkdir -p MyProject/src
199 mkdir -p MyProject/include
200 mkdir -p MyProject/bin
201 mkdir/docs
202 mkdir -p MyProject/docs
203 ls
204 tree MyProject
205 sudo apt install tree
206 tree MyProject
207 mkdir -p MyProject/src
208 tree MyProject
209 cd MyProject

```

```

210 git init
211 git status
212 nano src/hello.c
213 git status
214 nano src/hello.c
215 git status
216 ls src
217 cat src/hello.c
218 git add src/hello.c
219 git status
220 git commit -m "Added hello program"
221 git log
222 git commit -m "Added hello program"
223 git config --global user.email "himasreesurapuraju.com"
224 git config --global user.name "Himasree24"
225 git config --global user.email "himasreesurapuraju@gmail.com"
226 git config --global --list
227 git commit -m "Added hello program"
228 git log
229 git commit -m "Added hello program"
230 git push
231 git remote add origin https://github.com/Himasree24/Operating-Systems.git
232 git branch -M main
233 git push -u origin main
234 git remote add origin https://github.com/Himasree24/Operating-Systems.git
235 git branch -M main
236 git push -u origin main
237 git remote -v
238 git push -u origin main
239 git pull origin main --allow-unrelated-histories

```

```

240 git pull origin main --allow-unrelated-histories --no-rebase
241 git push -u origin main
242 pwd
243 cat exp1.c
244 gcc exp1.c -o exp1
245 cd ~/OSLab
246 cat exp1.c
247 ./exp1
248 cat exp2.c
249 ./exp2
250 cat exp3.c
251 ./exp3
252 cat exp4.c
253 ./exp4
254 cat exp5.c
255 ./exp5
256 cat exp6.c
257 ./exp6
258 cat exp7.c
259 ./exp7
260 cat exp8.c
261 ./exp8
262 cat exp9.c
263 ./exp9
264 ./exec
265 ~ ./exec
266 cd ~/exec
267 cd exec
268 cat programExec1.c
269 cat programExec1.c

```

```

270 ./programExec1
271 cat programExec2.c
272 ./programExec2
273 cat programExec3.c
274 ./programExec3
275 cat programExec4.c
276 ./programExec4
277 cat programExec5.c
278 ./programExec5
279 /OSLab
280 cd ~/OSLab
281 uname -a
282 lscpu
283 Lsblk
284 ps -ef
285 top
286 free -h
287 cd ~/OSLab
288 nano program1.c
289 cat program1.c
290 gcc program1.c -o program1
291 ./program1
292 nano program2.c
293 cat program2.c
294 gcc program2.c -o program2
295 ./program2
296 gcc program2.c -o program2
297 ./program2
298 nano program2.c
299 gcc program2.c -o program2

```

```

300 ./program2
301 gcc program2.c -o program2
302 ./program2
303 nano sample.txt
304 nano q2.c
305 nano sample.txt
306 cd ~/OSLab
307 nano sample.txt
308 gcc q2.c -o q2
309 ./q2
310 cat copy.txt
311 strace cat sample.txt
312 sudo apt update
313 sudo apt install strace
314 strace cat sample.txt
315 nano q2.c
316 nano program3.c
317 gcc program3.c -o program3
318 ./program3
319 nano program3.c
320 nano program4.c
321 gcc program4.c -o program4
322 ./program4
323 nano program4.c
324 nano form.c
325 gcc fork.c -o fork
326 gcc form.c -o form
327 ./form
328 nano form.c
329 nano excel.c

```

```

330 cd ~/OSLab
331 nano program4.c
332 nano excel.c
333 gcc excel.c -o excel
334 ./excel
335 nano q3.c
336 gcc q3.c -o q3
337 ./q3
338 nano q4.c
339 gcc q4.c -o q4
340 ./q4
341 nano q3.c
342 nano q4.c
343 nano q5.c
344 gcc q5.c -o q5
345 gcc q5.c -o q5
346 ./q5
347 pstree
348 nano q5.c
349 strace ls
350 strace -o trace.txt ls
351 cat trace.txt
352 nano shell.c
353 gc shell.c -o shell
354 ./shell
355 gcc shell.c -o shell
356 nano shell.c
357 gcc shell.c -o shell
358 ./shell
359 nano shell.c

```

```

360 cd ~/OSLab
361 nano program02.c
362 gcc program02.c -o program02
363 ./program02
364 ls -l | grep ".c"
365 nano program02.c
366 gcc program02.c -o program02
367 ./program02
368 nano program02.c
369 nano program2.3.c
370 gcc program02.3.c -o program02.3
371 gcc program2.3.c -o program2.3
372 ./program2.3
373 nano program23.c
374 gcc program23.c -o program23
375 ./program23
376 nano source.txt
377 ls
378 gcc program23.c -o program23
379 ./program23
380 nano program23.c
381 nano program24.c
382 gcc program24.c -o program24
383 ./program24
384 nano program24.c
385 nano program25.c
386 gcc program25.c -o program25
387 ./program25
388 nano program25.c
389 history

```